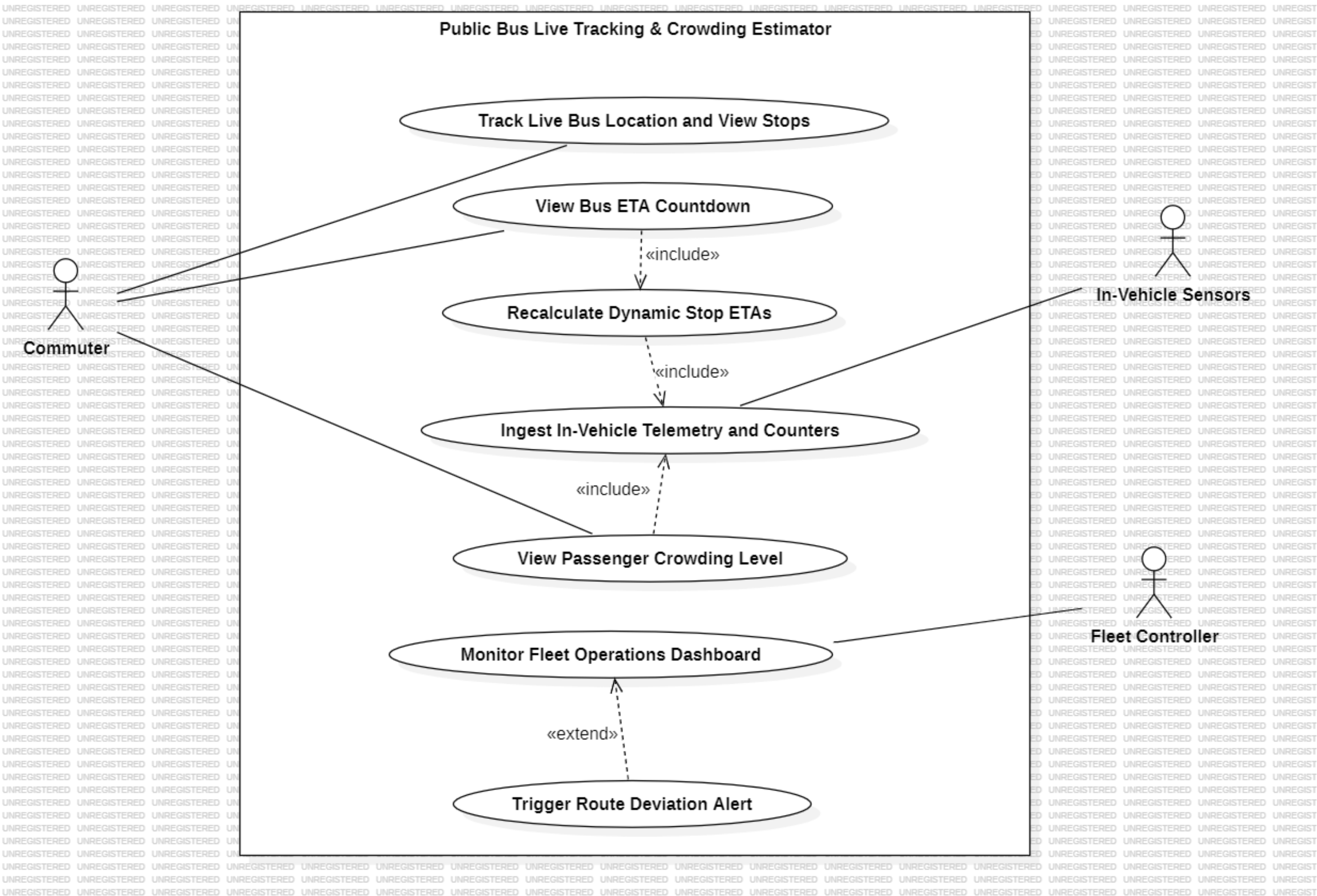
Software Engineering Lab 1

PES2UG24CS461

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria (measurable pass/fail)	Rationale	Comments (peer review)
FR-001	Functional	Input bus GPS coordinates every 5 seconds and recalculate arrival ETAs for all upcoming stops	High	**Pass:** ETA updates based on current speed and distance. **Fail:** ETA increases while bus approaches stop.	Accurate ETA is the main service for waiting commuters.	Peer: "Use standard 5s telemetry rate" - kept.
FR-002	Functional	Compute passenger crowding levels (Low, Medium, High) using ticketing validation sensor counts	High	**Pass:** Crowding level updates on UI within 3s of boarding. **Fail:** Count error exceeds 5% of actual load.	Helps commuters decide whether to board or wait for next bus.	Peer: "Add 3-tier crowding labels" - done.
FR-003	Functional	Provide a map view to search bus stops and view live moving bus positions	Medium	**Pass:** Selected stop shows live bus marker moving on map. **Fail:** Bus location deviates >50m from GPS feed.	Live visual tracking builds passenger trust in ETA times.	Peer: "Specify search and map view" - updated.
FR-004	Functional	Display a Fleet Controller dashboard showing real-time status and route adherence of all buses.	High	**Pass:** Dashboard lists all active buses with color-coded status. **Fail:** Telemetry list fails to refresh every 5s.	Fleet managers need full operational visibility to run buses on time.	Peer: "Ensure multi-bus view is stated" - added.
FR-005	Functional	Send an automated alert to the Fleet Controller if a bus deviates >200m from its assigned route.	Medium	**Pass:** Alert popup appears on dashboard within 10s of deviation. **Fail:** Alert fails to trigger on wrong turns.	Identifies route blocks, detours, and driver errors quickly.	Peer: "Why Medium?" - non-critical for core commuter view.
NFR-001	Nonfunctional (Performance & Security)	Tracking engine handles telemetry streams from up to 1,000 active transit vehicles simultaneously.	High	**Pass:** Latency remains <2s under simulated 1,000-bus load. **Fail:** Packet loss exceeds 0.1% during peak hours.	Prevents system slowdown during peak morning and evening traffic.	Peer: "Include exact vehicle capacity" - done.
NFR-002	Nonfunctional (Reliability)	The commuter API for live ETAs and crowding data shall maintain a 99.9% uptime per calendar month.	High	**Pass:** Server logs show <43.8 mins total unplanned downtime/month. **Fail:** Monthly downtime exceeds 44 mins.	Transit tracking must be available whenever commuters travel.	Peer: "Add measurable downtime minutes" - added.

UML Diagram – Public Bus Live Tracking & Crowding Estimator



Use Case Flow

Use Case: View Bus ETA Countdown

Primary Actor: Commuter

Secondary Actor: In-Vehicle Sensors (GPS & Ticketing)

Goal: View real-time arrival countdown and passenger crowding level for an incoming bus at a selected stop.

Trigger: Commuter selects a specific bus stop from the route map.

Includes: Recalculate Dynamic Stop ETAs (always runs to compute ETA based on speed and distance)

Preconditions

1. The commuter application is open with an active internet connection.
2. The transit bus is operational on the selected route and transmitting GPS telemetry every 5 seconds.
3. The selected bus stop ID is valid and active in the transit network database.

Postconditions

1. Commuter views the real-time arrival countdown (in minutes) for the selected bus.
2. Commuter views the current passenger crowding status (Low, Medium, High).
3. The system refreshes telemetry and ETA calculations every 5 seconds.

Main Success Scenario

1. Commuter searches for or taps a bus stop icon on the live transit map.
2. System retrieves all active buses currently assigned to routes serving that stop.
3. System fetches the latest GPS coordinates and passenger counts from In-Vehicle Sensors.
4. System executes **Recalculate Dynamic Stop ETAs** using current vehicle distance, speed, and intermediate stops.
5. System computes the crowding tier based on onboard ticketing count capacity.
6. System displays the incoming bus route number, arrival countdown (e.g., "5 mins"), and crowding badge (e.g., "Medium").
7. Commuter tracks the live incoming bus until arrival.
8. System updates coordinates and ETA countdown every 5 seconds.
9. Use case ends successfully.

Alternate Flow

Bus GPS telemetry signal is lost (dead zone / hardware outage)

1. System detects at step 4 that no GPS packet has been received for > 30 seconds.

2. System marks the live vehicle tracking status as "Offline / Stale".

3. System falls back to scheduled timetable data and displays the static arrival time accompanied by a warning badge: *"Scheduled time (Live GPS unavailable)"*.

4. System hides the real-time crowding indicator to prevent misleading commuters.

5. When fresh GPS packets resume, system automatically switches back to dynamic ETA countdown at step 6.